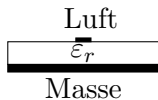


HF

1 Grundlagen

Mikrostreifenleitungen: Quasi-TEM-Wellen



Triplate Leitung: TEM-Wellen



Leitungsgleichung γ, Z_L

$$U = U_f e^{-\gamma z} + U_b e^{\gamma z}$$

$$I = \frac{U_f}{Z_L} e^{-\gamma z} - \frac{U_b}{Z_L} e^{\gamma z}$$

Normierungs:

$$u = \frac{U}{\sqrt{Z_L}}, i = I \sqrt{Z_0}, U_f = I_f Z_L$$

$$u = u_f e^{-\gamma z} + u_b e^{\gamma z} = a + b$$

$$i = a - b$$

Matrixbeschreibungen

- **S-Parameter (Streuparameter):**

- Beschreibung über Wellen (a_i (Eingang Oben), b_i (Ausgang unter), Nummer sind die Seite)

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

- * $S_{11}|_{a_2=0} = \frac{b_1}{a_1}$: Reflexion Eingang

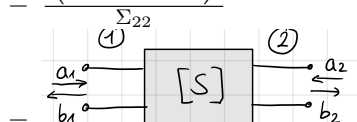
- * $S_{21}|_{a_2=0} = \frac{b_2}{a_1}$: Transmission (Verstärkung)

- * S_{12} : Rückwirkung

- * S_{22} : Reflexion Ausgang

$$\begin{pmatrix} A+B-C-D & 2\Delta_A \\ 2 & -A+B-C+D \end{pmatrix}$$

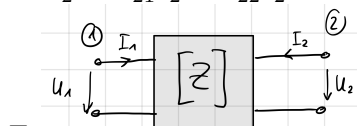
$$\begin{pmatrix} \Sigma_{12} & \Delta_{\Sigma} \\ 1 & -\Sigma_{21} \end{pmatrix}$$



- **Impedanzparameter:**

- $U_1 = Z_{11} I_1 + Z_{12} I_2$

- $U_2 = Z_{21} I_1 + Z_{22} I_2$

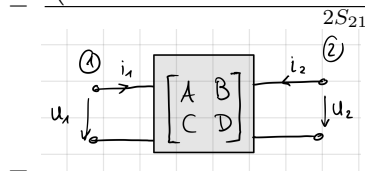


- **Kettenparameter:**

- Darstellung der ABCD-Parameter

$$\begin{pmatrix} U_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} U_2 \\ -I_2 \end{pmatrix}$$

$$\begin{pmatrix} -\Delta_S + S_{11} - S_{22} + 1 & \Delta_S + S_{11} + S_{22} + 1 \\ \Delta_S - S_{11} - S_{22} + 1 & -\Delta_S - S_{11} + S_{22} + 1 \end{pmatrix}$$



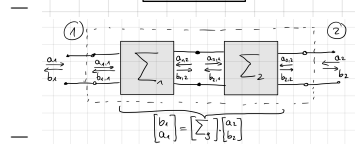
- **Transmissionsmatrix:**

- Multiplikation Vereinfachung

$$\begin{pmatrix} b_1 \\ a_1 \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$$

$$1 \begin{pmatrix} -\Delta_S & S_{11} \\ -S_{22} & 1 \end{pmatrix}$$

$$[\Sigma] = \frac{1}{S_{21}}$$



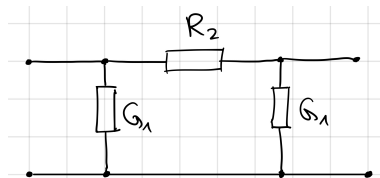
$$[\Sigma_g] = [\Sigma_1] [\Sigma_2]$$

1.1 Leitungen

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & \frac{R}{Z_0} \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ G Z_0 & 1 \end{pmatrix}$$

1.1.1 Dämpfungsglied



$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ G_1 Z_0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{R_2}{Z_0} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ G_1 Z_0 & 1 \end{pmatrix}$$

2 Koppler

Taktanregung

- **Gleichtaktanregung:** $\oplus$ Signale gleicher Amplitude und gleicher Phase.

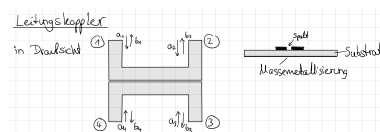
- Z_e
- $a_1^+ = \frac{1}{2} a_1, a_4^+ = \frac{1}{2} a_1$

- **Gegentaktanregung:** $\ominus$ Signale gleicher Amplitude und entgegengesetzter Phase.

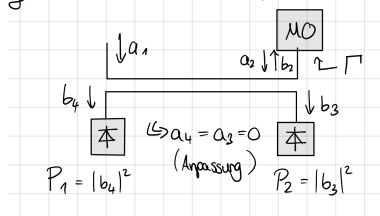
- Z_d
- $a_1^- = \frac{1}{2} a_1, a_4^- = -\frac{1}{2} a_1$

- Überlagerung: $a_1^+ + a_1^- = a_1, a_4^+ + a_4^- = 0$

2.1 Leitungskoppler

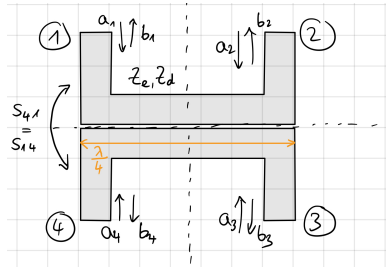


$$g_{es}: \Gamma = f([S], P_1, P_2)$$



$$|\Gamma|^2 = \frac{P_2}{P_1} \left| \frac{S_{41}}{S_{32} S_{21}} \right|^2$$

Zweifach Symmetrischer Zweitor:



Allseit. Anpassung:

$$S_{11} = S_{22} = S_{33} = S_{44} = 0$$

Symmetrien und Reziprok:

$$S_{41} = S_{14} = S_{32} = S_{23}$$

$$S_{12} = S_{21} = S_{34} = S_{43}$$

Mit Gleich- und Gegentaktanregung:

$$b_1 = b_1^+ + b_1^- = S_{11}^+ a_1^+ + S_{11}^- a_1^- = \frac{1}{2}(S_{11}^+ + S_{11}^-) a_1, \rightarrow S_{11}$$

$$b_2 = b_2^+ + b_2^- = S_{21}^+ a_1^+ + S_{21}^- a_1^- = \frac{1}{2}(S_{21}^+ + S_{21}^-) a_1, \rightarrow S_{21}$$

$$b_3 = b_2^+ + b_2^- = S_{21}^+ a_4^+ + S_{21}^- a_4^- = \frac{1}{2}(S_{21}^+ - S_{21}^-) a_1, \rightarrow S_{31}$$

$$b_4 = b_4^+ + b_4^- = S_{11}^+ a_4^+ + S_{11}^- a_4^- = \frac{1}{2}(S_{11}^+ - S_{11}^-) a_1, \rightarrow S_{41}$$

Normierung der Widerstände: $z_e = \frac{Z_e}{Z_0}, z_d = \frac{Z_d}{Z_0}$

$$\begin{bmatrix} A^+ & B^+ \\ C^+ & D^+ \end{bmatrix} = \begin{bmatrix} \cos(\beta l) & j z_e \sin(\beta l) \\ j \frac{1}{z_e} \sin(\beta l) & \cos(\beta l) \end{bmatrix}$$

$$S_{11}^+ = \frac{A^+ + B^+ - C^+ - D^+}{A^+ + B^+ + C^+ + D^+} = \frac{j z_e \sin(\beta l) - j \frac{1}{z_e} \sin(\beta l)}{2 \cos(\beta l) + j(z_e + \frac{1}{z_e}) \sin(\beta l)}$$

$$S_{11}^- = \frac{j(z_d - \frac{1}{z_d}) \sin(\beta l)}{2 \cos(\beta l) + j(z_d + \frac{1}{z_d}) \sin(\beta l)}$$

$$S_{21}^+ = \frac{2}{A^+ + B^+ + C^+ + D^+} = \frac{2}{2 \cos(\beta l) + j(z_e + \frac{1}{z_e}) \sin(\beta l)}$$

$$S_{21}^- = \frac{2}{A^- + B^- + C^- + D^-} = \frac{2}{2 \cos(\beta l) + j(z_d + \frac{1}{z_d}) \sin(\beta l)}$$

Allseitige anpassung

$$S_{11} = 0 \rightarrow S_{11}^+ = -S_{11}^- \rightarrow z_e = \frac{1}{z_d} \leftrightarrow z_e z_d = 1 \rightarrow Z_e Z_d = Z_0^2$$

$$\text{Entkoppelte Tore: } S_{31} = \frac{1}{2}(S_{21}^+ - S_{21}^-) = 0 \rightarrow S_{21}^+ = S_{21}^-$$

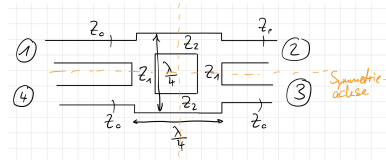
$$S_{21} = S_{21}^+ = \frac{2}{2 \cos(\beta l) + j(z_e + \frac{1}{z_e}) \sin(\beta l)}$$

$$S_{41} = \frac{1}{2}(S_{11}^+ - S_{11}^-) = S_{11}^+ = \frac{j(z_e - \frac{1}{z_e}) \sin(\beta l)}{2 \cos(\beta l) + j(z_e + \frac{1}{z_e}) \sin(\beta l)}$$

$$\angle(S_{21}, S_{41}) = 90^\circ$$

Mehrstufig wenn höhere Breitband.

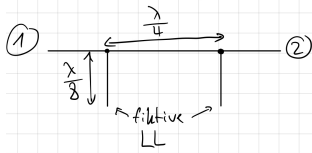
2.2 Ringkoppler



- 3dB-Kopplung: $1 \rightarrow 2$ und $1 \rightarrow 3$
- Entkopplung: 1 und 4
- Einspeisung an 1
- $Z_1 = ?$, $Z_2 = ?$

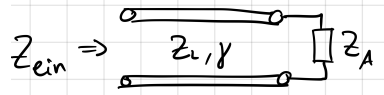
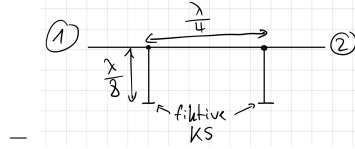
• Gleichtakt

$$- \oplus \text{ an 1 und 4: } a_1^+ = \frac{1}{2} a_1, a_4^+ = \frac{1}{2} a_1$$



• Gegentakt

$$- \ominus \text{ an 1 und 4: } a_1^- = \frac{1}{2} a_1, a_4^- = -\frac{1}{2} a_1$$



- $Z_{ein} = Z_L \frac{Z_A + j Z_L \tan(\beta l)}{Z_L + j Z_A \tan(\beta l)}$
- $\tan(\beta l) = \tan(\frac{2\pi}{\lambda} \frac{\lambda}{8}) = \tan(\frac{\pi}{4}) = 1$
- $\oplus Z_A \rightarrow \infty \Rightarrow Z_{ein,LL}^+ = \lim_{Z_A \rightarrow \infty} Z_{ein} = -j Z_L = -j Z_1$
- $\ominus Z_A \rightarrow 0 \Rightarrow Z_{ein,LL}^- = j Z_L = j Z_1 = j \frac{1}{Y_1}$

Kettenmatrizen fortsetzung

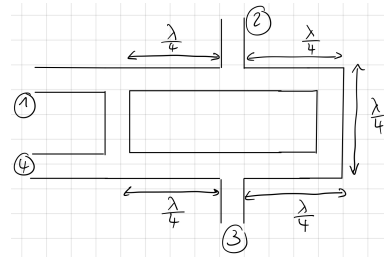
$$\begin{bmatrix} M^+ \\ M^- \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -j Y_1 Z_0 & 1 \end{bmatrix} \begin{bmatrix} 0 & j \frac{Z_2}{Z_0} \\ j \frac{Z_0}{Z_2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -j Y_1 Z_0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} +Z_2 Y_1 & j \frac{Z_2}{Z_0} \\ j \frac{Z_0}{Z_2} - j Y_1^2 Z_0 Z_2 & +Y_1 Z_2 \end{bmatrix} = \begin{bmatrix} A^+ & B^+ \\ C^+ & D^+ \end{bmatrix}$$

Anforderung: Allseitige Anpassung

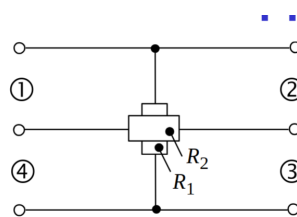
- $S_{11} = \frac{1}{2}(S_{11}^+ + S_{11}^-) \stackrel{!}{=} 0$
- $S_{41} = S_{14} = \frac{1}{2}(S_{11}^+ - S_{11}^-) \stackrel{!}{=} 0$
- $S_{11}^+ = \frac{A^+ + B^+ - C^+ - D^+}{A^+ + B^+ + C^+ + D^+} \stackrel{!}{=} 0$
- Da $A^+ = D^+ \Rightarrow B^+ = C^+$
- $\frac{Z_2}{Z_0} = \frac{Z_0}{Z_2} - Y_1^2 Z_2 Z_0$
- $|S_{21}| = |\frac{1}{2}(S_{21}^+ + S_{21}^-)| = |S_{31}| = |\frac{1}{2}(S_{21}^+ - S_{21}^-)|$
- $\frac{Z_2}{Z_0} = Y_1 Z_2 \Rightarrow Y_1 = \frac{1}{Z_0}$
- $1 = \frac{Z_0^2}{Z_2^2} - Y_1^2 Z_0^2 \Rightarrow Z_2 = \frac{1}{\sqrt{2}} Z_0$
- $\angle(S_{21}, S_{31}) = 90^\circ$
- 3dB-Kopplung: $1 \rightarrow 2$ und $1 \rightarrow 3$
- Entkopplung: 1 und 4

2.2.1 Rat-Race-Koppler



- Einsp. 1: 3 entkoppelt, $\angle(S_{21}, S_{41}) = 0^\circ$
- Einsp. 3: 1 entkoppelt, $\angle(S_{23}, S_{43}) = 180^\circ$

2.3 Resistiver Koppler

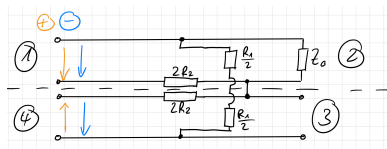


Synthese: All. Anpassung

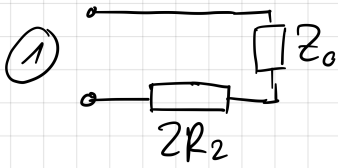
- $S_{31} = S_{13} = S_{24} = S_{42} = 0$

$$S_{11} = \frac{Z_{ein} - Z_0}{Z_{ein} + Z_0}$$

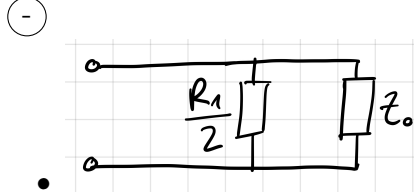
$$\oplus \text{ und } \ominus \text{ an 1 und 4.}$$



$\oplus \Rightarrow$ in der Symmetrieeben $\Rightarrow$ LL



$$S_{11}^+ = \frac{2R_2 + Z_0 - Z_0}{2R_2 + Z_0 + Z_0} = \frac{R_2}{R_2 + Z_0}$$



$$S_{11}^- = -\frac{Z_0}{R_1 + Z_0}$$

$$S_{11} = \frac{1}{2}(S_{11}^+ + S_{11}^-) \stackrel{!}{=} 0 \Rightarrow S_{11}^+ = -S_{11}^-$$

$$\frac{R_2}{R_2 + Z_0} = \frac{Z_0}{R_1 + Z_0} \Rightarrow R_1 R_2 = Z_0^2$$

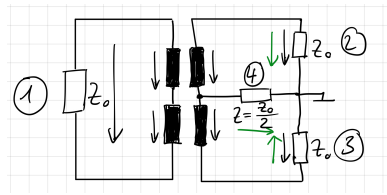
$$S_{41} = \frac{1}{2}(S_{11}^- - S_{11}^+) = S_{11}^+ = \frac{R_2}{R_2 + Z_0}$$

$$S_{21} = \frac{Z_0}{Z_0 + R_2}$$

Häufige Wahl: $R_1 = R_2 = Z_0 \Rightarrow S_{21} = S_{41} = \frac{1}{2}$

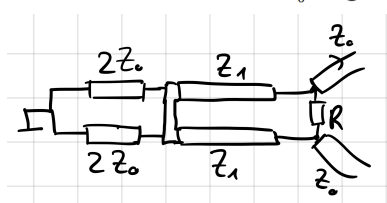
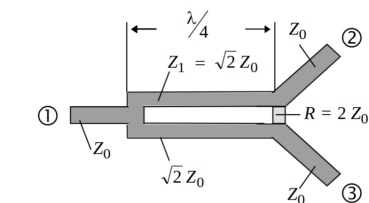
Nullgradkoppler: $\angle(S_{21}, S_{41}) = 0^\circ$

2.4 Transformatorkoppler



- Transformator sei ideal
- aus Symmetrie sind 1 und 4 entkoppelt
- bei Einspeisung an 1 $\Rightarrow$ Signale an 2 und 3 in Gegenphase
- bei Einspeisung an 4 $\Rightarrow$ Signale an 2 und 3 in Phase
- $\Rightarrow$ Signale an 2 und 3 entkoppelt mit $Z = \frac{Z_0}{2}$

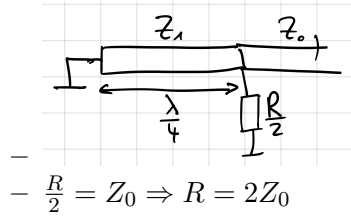
2.5 Nullgradkoppler/Signalteiler



- $\oplus$ und $\ominus$ an 2 und 3
- $\oplus$ an 2 und 3:

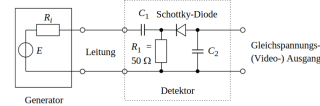
$$Z_{ein} \frac{Z_1}{2Z_0} \stackrel{!}{=} 0 \Rightarrow Z_1 = \sqrt{2}Z_0$$

- $\ominus$ an 2 und 3:

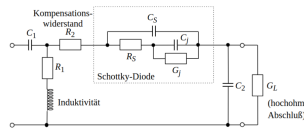


3 Messung

3.1 Breitbanddetektoren



Detektor mit Zwangsanpassungswiderstand R_1



Nicht Lineare

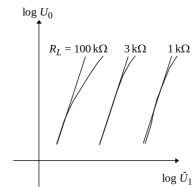
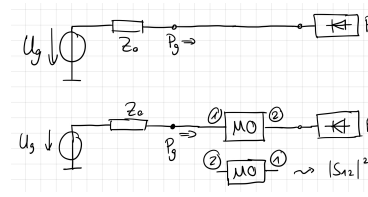


Bild 1.27: Detektorausgangsspannung U_0 als Funktion des HF-Pegels $\hat{U}_1$ für verschiedene Lastwiderstände R_L

Effekte (Verständnis)

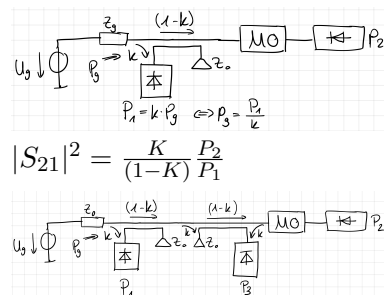
3.2 Skalare Zweitormessung

3.2.1 Substitutionsverfahren



$$P_2 = |S_{21}|^2 P_g \Leftrightarrow |S_{21}|^2 = \frac{P_2}{P_1}$$

3.2.2 Leitungskoppler

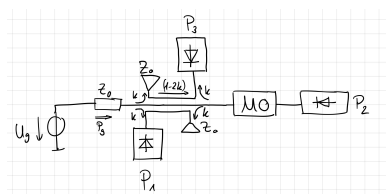


$$P_1 = K P_g$$

$$|S_{21}|^2 = \frac{K}{(1-K)} \frac{P_2}{P_1}$$

- $P_2 = |S_{21}|^2 (1-k)^2 P_g$
- $P_1 = k P_g$
- $P_3 = |S_{11}|^2 (1-k)^2 k P_g$
- $|S_{21}|^2 = \frac{P_2}{P_1} \frac{k}{(1-k)^2}$
- $|S_{11}|^2 = \frac{P_3}{P_1} \frac{1}{(1-k)^2}$

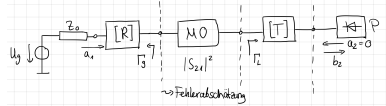
3.2.3



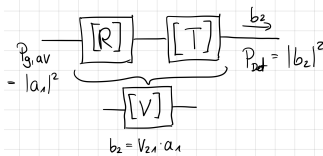
- $P_1 = kP_g$
- $P_2 = |S_{21}|^2(1 - 2k)P_g$
- $P_3 = k|S_{11}|^2(1 - 2k)P_g$
- $|S_{21}|^2 = \frac{P_2}{P_1} \frac{k}{1-2k}$
- $|S_{11}|^2 = \frac{P_3}{P_1} \frac{1}{1-2k}$

3.2.4 Fehler bei Transmissionmessung

Annahme: Fehlanpassung am Generator und am Leistungsdetektor → Fehlermodell

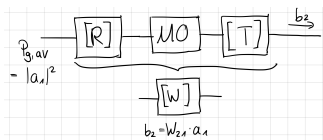


- $P_{Det} = |b_2|^2$
- Ohne MO:



- $b_2 = V_{21} \cdot a_1$
- $P_{gver} = |a_1|^2$
- $b_2 = V_{21}a_1$
- $P_{Det} = |b_1|^2 = |V_{21}|^2|a_1|^2$
- $R_{22} = \Gamma_g$
- $T_{11} = \Gamma_L$
- Zu Transmissionmatrizen
- $[\Sigma_V] = [\Sigma_R] [\Sigma_T] =$
 $\frac{1}{R_{21}} \begin{bmatrix} -\Delta_R & R_{11} \\ -R_{22} & 1 \end{bmatrix} \frac{1}{T_{21}} \begin{bmatrix} -\Delta_T & T_{11} \\ -T_{22} & 1 \end{bmatrix}$
- $V_{21} = \frac{1}{\Sigma V_{22}} = \frac{R_{21}T_{21}}{1-R_{22}T_{11}}$
- $P_{Det} = \left| \frac{R_{21}T_{21}}{1-R_{22}T_{11}} \right|^2 P_{gver}$
- Annahme: Fehlerzweitoren verlustlos:
 - * $|R_{21}|^2 = |R_{12}|^2 = 1 - |R_{22}|^2$
 - * $|T_{21}|^2 = 1 - |T_{11}|^2$
- $P_{Det} = \frac{(1-|\Gamma_g|^2)(1-|\Gamma_L|^2)}{|1-\Gamma_g\Gamma_L|^2} P_{gver}$

- Mit MO:



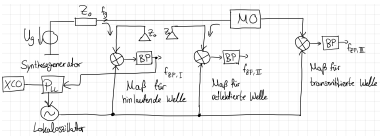
- $b_2 = W_{21} \cdot a_1$
- $W_{21} = \frac{1}{\Sigma W_{22}}$
- $\Sigma W =$
 $\frac{1}{R_{21}S_{21}T_{21}} \begin{bmatrix} -\Delta_R & R_{11} \\ -R_{22} & 1 \end{bmatrix} \begin{bmatrix} -\Delta_S & S_{11} \\ -S_{22} & 1 \end{bmatrix} \begin{bmatrix} -\Delta_T & T_{11} \\ -T_{22} & 1 \end{bmatrix}$
- $W_{21} = \frac{R_{21}S_{21}T_{21}}{1-R_{22}S_{11}-S_{22}T_{11}+T_{11}R_{22}\Delta_S}$
- $|\hat{S}_{21}|^2 = \frac{|W_{21}|^2}{|V_{21}|^2} =$
 $|S_{21}|^2 \frac{|1-\Gamma_g\Gamma_L|^2}{|(1-\Gamma_gS_{11})(1-\Gamma_LS_{22})-S_{12}S_{21}\Gamma_g\Gamma_L|^2}$

- Beispiel für Fehlerschätzung:

- MO: $|S_{12}| = |S_{21}| = \frac{1}{2}$, $|S_{11}| = |S_{22}| = 0.1$
- Messsystem: $|\Gamma_g| = 0.1$, $|\Gamma_L| = 0.1$

$$\begin{aligned} - \left| \hat{S}_{21} \right|^2 &= \frac{|1 \pm 0.01|^2}{|(1 \mp 0.01)^2 \mp 0.0025|^2} |S_{21}|^2 \\ - \left| \hat{S}_{21} \right|^2 &\approx \frac{1 \pm 0.02}{(1 \mp 0.02 \mp 0.0025)^2} |S_{21}|^2 \\ - \left| \hat{S}_{21} \right|^2 &\approx \frac{1 \pm 0.02}{1 \mp 0.045} |S_{21}|^2 \approx (1 \pm 0.06) |S_{21}|^2 \hat{=} 0.27dB \end{aligned}$$

3.3 Vektorielle Netzwerkanalyse



- Maß für fortlaufende Welle $f_{ZF,I}$
- Maß für reflektierte Welle $f_{ZF,II}$
- Maß für transmittierte Welle $f_{ZF,III}$
- **Fehlt Bild**(f_g, S_{21})